Bachelor in Artificial Intelligence

AI for Communication and Marketing

LAB EXAM
PART 2

Customer Churn Prediction
& Retention Strategy

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16.8% of 5,628 customers are churners → standard accuracy is misleading, **Recall** is the metric that matters.

Tenure

#1 RISK DRIVER

32%
churn in first 6 months

New customers (0-6 months) churn at 32%: 3x more than those with 7-12 months. Beyond 25 months, churn drops to 0% across 429 customers

35% Feature Importance
-0.34 Correlation

Complaint Filed

#2 RISK DRIVER

31.7%
churn rate with complaint

Customers who filed a complaint churn at 31.7% vs 10.9% who did not, nearly 3x higher. The gap persists across every satisfaction score level

8% Feature Importance
+0.25 Correlation

Cashback Amount

#3 RISK DRIVER

10%
feature importance

Higher cashback is consistently associated with lower churn. Unlike Tenure, it is a directly controllable marketing lever

10% Feature Importance
-0.15 Correlation

→ Customers who are **new, unheard and undervalued** are the ones who leave

Model: **Random Forest Tuned · Recall 0.874 · Lift 5.61x**
Target segment: **Tenure ⩽ 6 months + top 20% churn score** → reaches 3.1× more churners than random

Addresses Tenure

Onboarding Sequence

Personalized touchpoints at day 7, 30, 90, 180 – milestone celebrations and product recommendations based on PreferedOrderCat

Why this works:

Tenure is the dominant predictor. The KDE analysis shows churners are concentrated near 0 months. Building engagement before the customer disengages closes the window of maximum risk.

Addresses Complaint Filed

Complaint Fast-Track

Any new customer (Tenure ⩽ 6 months) who files a complaint is escalated to a retention agent within 24 hours

Why this works:

Customers who complained churn at 31.7% — nearly 3× those who did not. The interaction analysis confirms this holds across all satisfaction score levels. Speed of response is the critical variable.

Addresses Cashback Amount

Cashback Boost

+15% cashback on next 3 orders for all customers with Tenure ⩽ 6 months

Why this works:

Customers who complained churn at 31.7% — nearly 3× those who did not. The interaction analysis confirms this holds across all satisfaction score levels. Speed of response is the critical variable.